Introduction to Amazon Web Service(AWS)

(AWS) is a cloud computing platform provided by Amazon. It offers on-demand IT services like servers, storage, databases, networking, and more — all delivered over the internet.

Instead of buying physical servers, companies can rent resources from AWS and pay only for what they use.

🔹 **Common AWS Services:**

- **EC2 (Elastic Compute Cloud)** – Virtual servers
- **S3 (Simple Storage Service)** – Cloud storage
- **RDS (Relational Database Service)** – Managed databases
- **Lambda** – Serverless computing
- **VPC (Virtual Private Cloud)** – Networking service

✅ Advantages of AWS

1️⃣ Cost Effective

- Pay-as-you-go pricing model
- No need to buy physical hardware

2️⃣ Scalability

- Easily increase or decrease resources
- Auto-scaling available

3️⃣ Global Infrastructure

- Data centers available worldwide

4️⃣ High Availability

- Services are reliable and fault-tolerant

5️⃣ Security

- Strong security features

- IAM (Identity and Access Management) for access control

6️⃣ **Large Service Ecosystem**

- 200+ cloud services available

❌ **Disadvantages of AWS**

1️⃣ **Complex Pricing**

- Pricing structure can be confusing

2️⃣ **Learning Curve**

- Many services → difficult for beginners

3️⃣ **Dependency on Internet**

- Requires stable internet connection

4️⃣ **Vendor Lock-in**

- Difficult to migrate applications to another cloud provider

5️⃣ **Technical Support Cost**

- Premium support plans are expensive

AWS Regions

An **AWS Region** is a physical geographic location in the world where AWS has multiple data centers.

Each region contains multiple **Availability Zones (AZs)**.

Example AWS Regions:

- Mumbai (ap-south-1)
- US East (N. Virginia)
- London (eu-west-2)
- Singapore (ap-southeast-1)

🔹 What is Availability Zone (AZ)?

An **Availability Zone** is one or more data centers inside a region.

Example:
If you choose the **Mumbai Region**, it may have:

- ap-south-1a
- ap-south-1b
- ap-south-1c

Each AZ is isolated but connected to others for high availability.

🎯 Why Regions Are Important?

- Reduce latency (choose region close to users)
- Data compliance requirements
- Disaster recovery setup
- High availability architecture

🔁 Simple Example

If your users are in **India**, choose:
👉 Mumbai Region (ap-south-1)

If your users are in **USA**, choose:
👉 US East (N. Virginia)

This improves performance and reduces delay.

If you want, I can also explain:

- Difference between Region and Availability Zone
- AWS Global Infrastructure diagram explanation

- AWS Free Tier details
- AWS interview questions & answers 😊